A Where the background comes from

Read-level test (e.g. LoFreq, Geneious)



each base is judged by its own quality score, so errors that recur at one position (PCR, homopolymers, paralog reads) pass as mutations, and reads from one animal are counted as if independent

✓ correct decision ✕ wrong decision

Control-calibrated test (SHMtrace)



error rate per substitution, strand and base quality, measured in matched AID-null libraries from the same run; a call needs support beyond it (BH FDR ≤ 0.05); groups are compared one value per animal

B Inputs and read preparation



Nanopore
full-length amplicons



Sanger
.ab1 traces + sample sheet



Illumina
paired reads, merged



UMI-tagged
one consensus per molecule

Read preparation

1. **Length and expected-error gates**
   over-length reads and reads averaging > 0.01 errors per base
2. **Competitive alignment**
   minimap2 against target and paralog references; chimeras out
3. **Paralog-allele filter**
   reads with ≥ 10 paralog-specific bases removed
4. **Matched depth**
   random draw of the same number of usable reads per library

C Calling and quality checks

1 Control error model
per substitution, strand and base quality; a margin chosen by leave-one-control-out

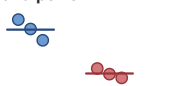
2 Test and FDR
support against expected errors; Benjamini–Hochberg FDR ≤ 0.05; each control called vs the others

3 Germline and window
sites at ≥ 50% removed; calls restricted to the covered window; indels optional

4 Automatic checks
spectrum (C→T/G→A share), positional clustering, 5-mer targeting index, Sanger bookkeeping flags

D Outputs

Animal-level tests and power



NB, Welch, permutation

Single-read estimate



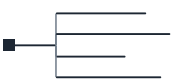
observed minus control background

Clones and clone-weighted rate



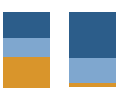
V(D)J junction as a barcode

Mutation trees



full-length reads of one clone

Repair-route indices



G:C Ts / Tv, A:T, WA/TW, decomposition

Reports, app and command line



HTML report, tables, project files